Binary EFD formula integration

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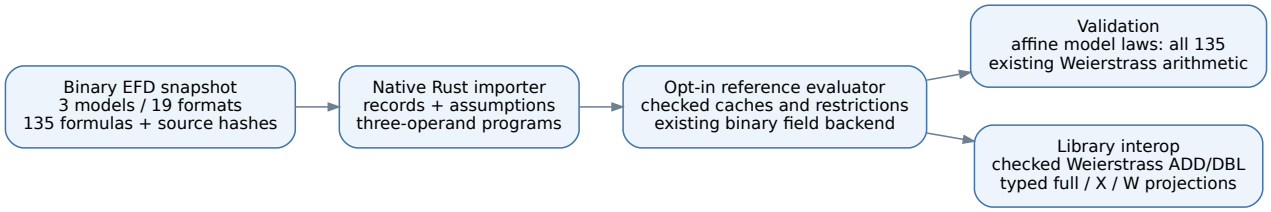
The linked characteristic-two Explicit-Formulas Database section is imported as 135 executable reference programs in 19 representations on three models. The source snapshot includes assumptions, curve and coordinate definitions, attribution, URLs and SHA-256 hashes. The methods are established prior work.



Delivered scope

The library exposes finite-point encoding, typed full/X/W projection, program compilation and evaluation, and checked Weierstrass addition/doubling adapters. Parameter and mixed-input restrictions are checked. Exceptional full-point operations use the existing affine law and expose their execution path. Both repositories contain identical core source, catalogue and native tooling.

Operation	Imported programs
Addition	41
Doubling	47
Tripling	2
Differential addition	16
Ladder step	16
Scaling	13

Validation

All 135 programs agree with the unoptimized affine model laws over GF(2^4) and GF(2^5): 8,640 successful formula/input comparisons per repository. Unit and nonunit scales are exercised where allowed; singular affine charts and zero denominators are counted separately in coverage.json. Applicable full-point Weierstrass operations also agree with pre-existing arithmetic, including public known-scalar fixtures at 113, 131 and 163 bits. Source hashes, cache and parameter mismatches, infinity, opposites, missing charts and field-width errors have explicit tests.

These are finite correctness checks, not universal proofs. The upstream Sage scripts are linked but were not replayed. See REPORT.md and validation/ for exact commands, host and complete execution status, including wider suites.

Limits

This import covers the linked binary-field section, not prime-field formulas. Low-level callers establish model-point membership and differential relations. Curve model changes are explicit. Production rho walks and GPU kernels are unchanged; selectors must remain independent of projective scaling. No speedup or new cryptanalytic result is claimed. GPU tuning and complete-step timings remain unmeasured. Existing result graphs have no new performance value to plot.

Sources

Bernstein–Lange binary EFD index lambda formulas the three imported model definitions; crypto:ecc2k130/LAMBDA-PROJECTIVE.md. The full source inventory and requirement-to-evidence table are in README.md and REPORT.md beside this PDF.